

ASSIGNMENT -1



K. R. MANGALAM UNIVERSITY

BTech CSE (AI & ML)

Section - B

SOET (School of Engineering and Technology)

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COURSE: Computer Networks

Course Code: (ENCS352)

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Assignment 1: Basic Network Topologies Using Switches and Hubs

Experiment Objectives:

In this experiment, students will create three small LAN designs in Cisco Packet Tracer to understand how network topology influences connectivity, fault tolerance, and performance. Students will configure end devices with IPv4 addressing and verify communication using ICMP.

Learning Outcomes

- Create small networks in Packet Tracer using different topologies
- Assign IP addresses correctly and test connectivity
- Identify failure points in different physical layouts
- Compare topology behaviour using simulation results

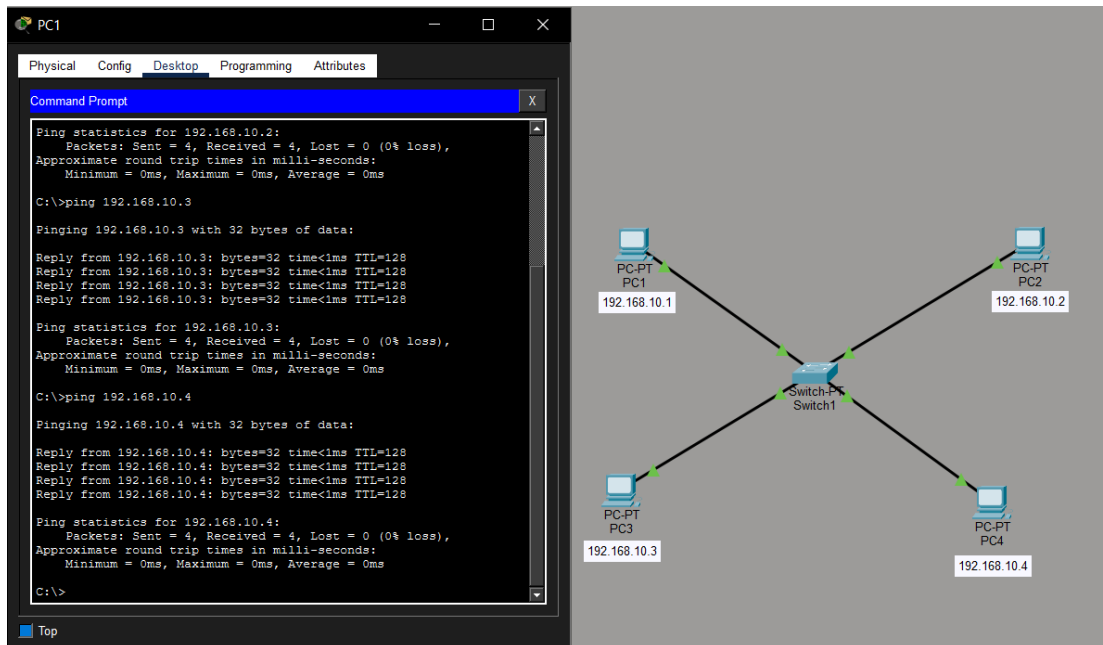
Concepts Used

- Star / Hub-based / Loop (ring-like) layouts
- IPv4 addressing and subnetting
- ICMP ping
- Packet Tracer Simulation Mode

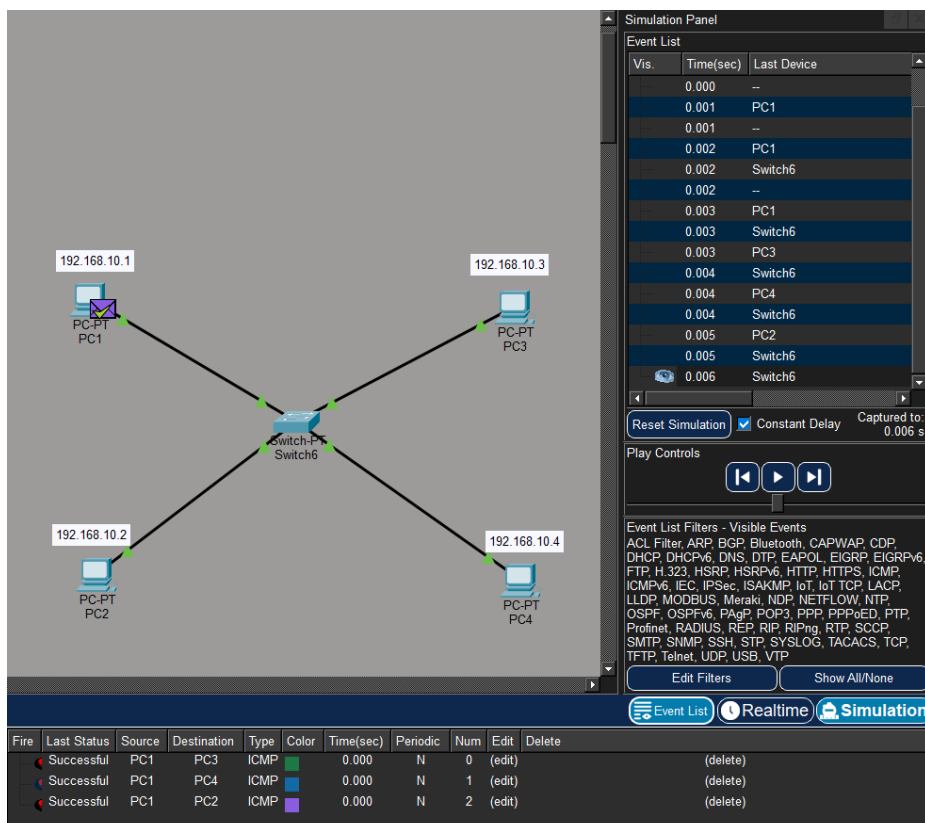
GITHUB LINK: [Nik848/CN_LAB_Assignments_KRMU](https://github.com/Nik848/CN_LAB_Assignments_KRMU)

Task 1: Star Topology (Switch)

- Place 1 switch + 4 PCs
- Connect all PCs to the switch
- Assign IPs (example): 192.168.10.1–4 /24
- Ping PC1 → PC2, PC3, PC4



Simulation:



Behavior

All PCs are connected to a central switch and configured in the same subnet (192.168.10.0/24). Ping results show successful replies with 0% packet loss. The switch forwards frames only to the intended destination using its MAC address table. Communication is fast and efficient with minimal delay. The simulation confirms proper ICMP packet delivery.

Failure Behavior

If one PC cable is disconnected, only that PC loses connectivity while the rest of the network continues to function. However, if the central switch fails, the entire network communication stops. The switch acts as a single point of failure. This topology provides partial fault isolation but depends heavily on the switch.

Practical Applications (LAN Design)

Star topology is widely used in offices, schools, and labs. It provides better performance and easier troubleshooting compared to bus topology. Network expansion is simple by adding more ports or switches. It supports modern Ethernet networks and structured cabling systems.

Conclusion

The star topology using a switch ensures reliable and efficient communication. It reduces collisions and improves overall performance. Although the switch is a central dependency, it provides better control and scalability. This topology is suitable for most real-world LAN implementations.

Task 2: Bus-like Topology (Hub)

- Replace switch with hub
- Keep same IP scheme
- Send pings simultaneously and observe packet flow

The screenshot displays a network simulation interface. On the left, a 'Command Prompt' window on PC1 shows the following output:

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=47ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 47ms, Average = 11ms

C:\>
```

On the right, the network diagram shows a central 'Hub-PT Hub1' connected to four PCs: PC-PT PC1 (192.168.10.1), PC-PT PC2 (192.168.10.2), PC-PT PC3 (192.168.10.3), and PC-PT PC4 (192.168.10.4). All connections are bidirectional.

Simulation:

The screenshot displays a network simulation interface. On the left, the network diagram shows a central 'Hub-PT Hub1' connected to four PCs: PC-PT PC1 (192.168.10.1), PC-PT PC2 (192.168.10.2), PC-PT PC3 (192.168.10.3), and PC-PT PC4 (192.168.10.4). All connections are bidirectional.

On the right, the 'Simulation Panel' is visible, showing an 'Event List' with the following data:

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC1
	0.002	Hub1
	0.002	Hub1
	0.002	Hub1
	0.003	PC4
	0.004	Hub1
	0.004	Hub1
	0.004	Hub1

Below the event list, there are 'Play Controls' (Reset Simulation, Constant Delay, Captured to...) and 'Event List Filters - Visible Events' (ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TOP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, PDR3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP). At the bottom, there are buttons for 'Edit Filters' and 'Show All/None'.

At the bottom of the simulation panel, there is a table with the following data:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC4	ICMP		0.000	N	0	(edit)	(delete)

Behavior

All PCs are connected to a central hub using the same IP subnet (192.168.10.0/24). Ping results are successful with replies received from all destination PCs. In simulation mode, packets are broadcast to all connected devices before reaching the intended recipient. The hub does not filter traffic; it forwards data to every port. Communication works correctly but is less efficient compared to a switch.

Failure Behavior

If one PC cable is disconnected, only that device loses connectivity while others continue communicating. However, if the hub fails, the entire network stops functioning. Since a hub broadcasts traffic, collisions may occur when multiple devices transmit simultaneously. This can reduce performance and increase delay under heavy traffic conditions.

Practical Applications (LAN Design)

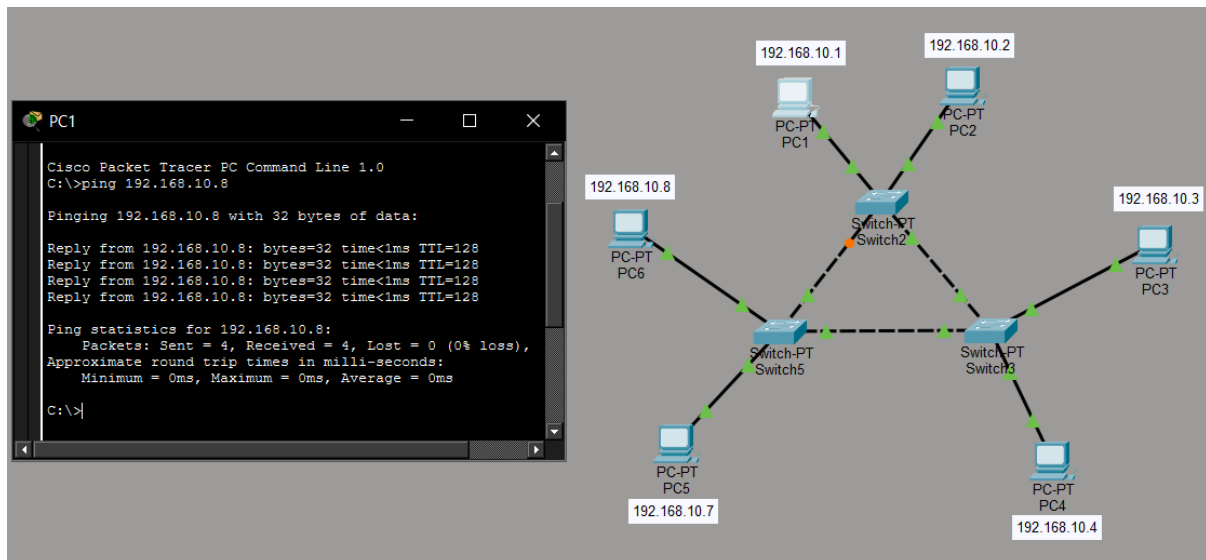
Hub-based topology is inexpensive and simple to set up. It was commonly used in early Ethernet LANs and small temporary networks. However, due to poor performance and high collision rates, hubs are rarely used in modern networks. Switches have largely replaced hubs in real-world LAN design.

Conclusion

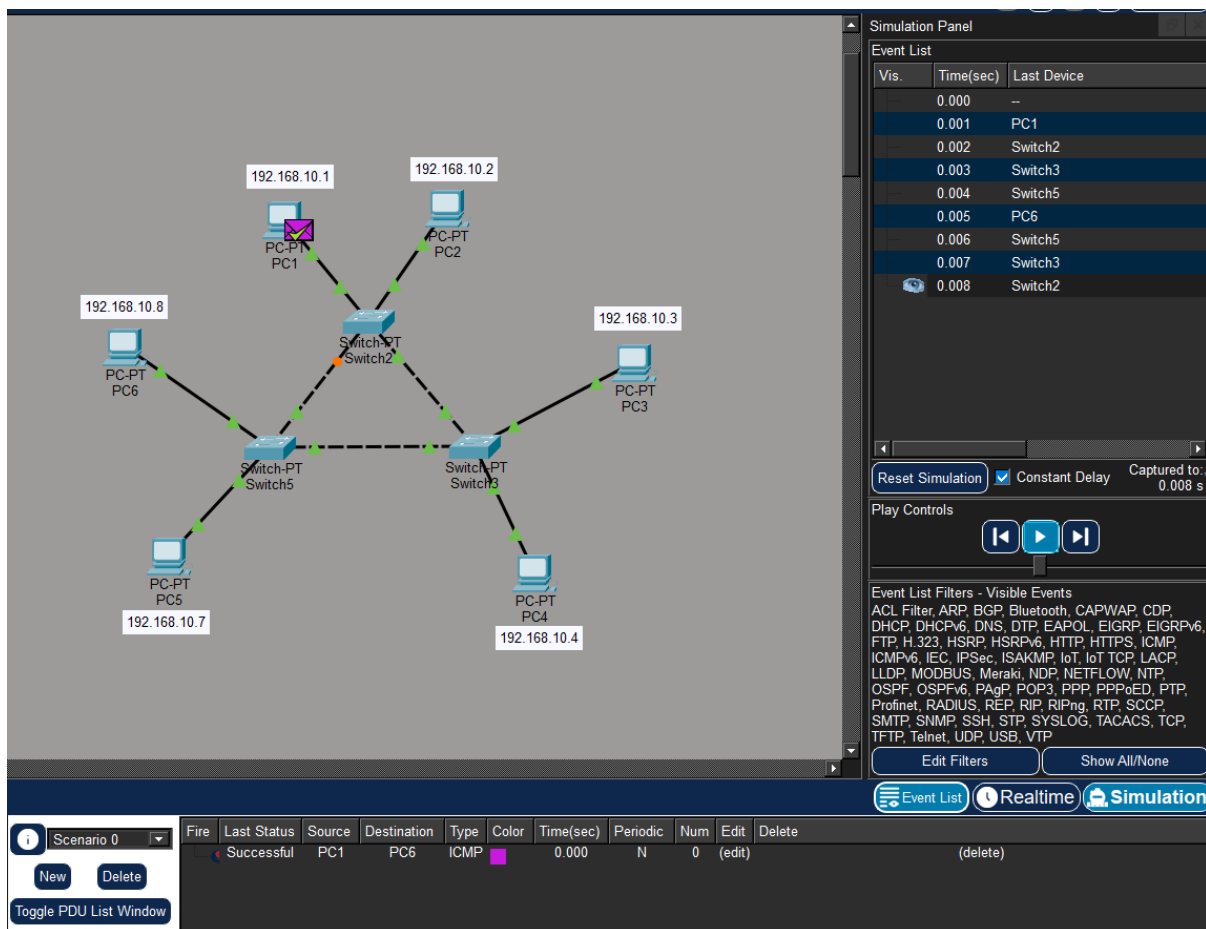
The hub topology allows successful communication within the same subnet but operates less efficiently than a switch. It broadcasts traffic to all devices, increasing collision chances. While simple and low-cost, it lacks scalability and performance optimization. Therefore, it is suitable only for basic learning purposes and small networks.

Task 3: Ring-like Topology (Loop)

- Place 3 switches and connect them in a loop
- Attach 1–2 PCs to each switch
- Ping across switches and observe path



Simulation:



Behavior

Multiple switches are connected in a loop structure with PCs attached to each switch. All devices are configured in the same subnet (192.168.10.0/24), and ping results are successful with 0% packet loss. In simulation mode, packets travel across multiple switches before reaching the destination. The network maintains connectivity across interconnected switches. Communication works efficiently due to switching at each device.

Failure Behavior

If one link in the ring is disconnected, traffic can still reach the destination through the alternate path. This provides redundancy and improves fault tolerance. However, if multiple links fail, communication between certain segments may stop. Without proper loop control protocols, broadcast storms could occur in real networks.

Practical Applications (LAN Design)

Ring-like or looped switch designs are used in enterprise networks for redundancy. They improve reliability by providing multiple paths for data transmission. This topology is useful in campuses, data centers, and large office networks. It enhances fault tolerance compared to simple star or hub layouts.

Conclusion

The ring-like topology provides better reliability and path redundancy. Communication remains stable even if one link fails. It offers improved fault tolerance and scalability. Therefore, it is more suitable for larger and more reliable network designs.

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- The screenshot displays a Cisco Packet Tracer PC Command Line window for PC1 (192.168.10.1) and a corresponding network diagram. The command prompt shows a successful ping from PC1 to PC2 (192.168.10.6).
- Command Prompt Output:**
- ```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.6:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```
- Network Diagram:**
- The network diagram shows a multi-switch topology with six PCs connected to three switches (Switch-P1, Switch-P2, Switch-P3). The connections are as follows:
- Switch-P1 is connected to PC1 (192.168.10.1) and PC2 (192.168.10.2).
  - Switch-P2 is connected to PC3 (192.168.10.3) and PC4 (192.168.10.4).
  - Switch-P3 is connected to PC5 (192.168.10.5) and PC6 (192.168.10.6).
- The switches are interconnected in a mesh topology, with Switch-P1 connected to Switch-P2 and Switch-P3, and Switch-P2 connected to Switch-P3.

## **Behaviour/Observation**

After disconnecting one link in the ring-like topology, communication between PCs is still successful. Ping results show replies with 0% packet loss. The network automatically uses the alternate path to reach the destination. Simulation mode confirms that packets are rerouted through another switch. This demonstrates redundancy in the design.

## **Failure Behavior**

When a single link fails, the network continues operating due to the loop structure. However, if more than one critical link fails, certain segments may become isolated. The topology reduces the impact of a single point of failure. It provides higher fault tolerance compared to star or hub designs.

## **Practical Applications (LAN Design)**

This type of redundant design is used in enterprise and campus networks where reliability is important. It ensures continuous operation even during cable or link failures. Such designs are common in backbone networks and data centers. Redundancy improves network availability and stability.

## **Conclusion**

The failure test confirms that ring-like topology offers better resilience. Even after disconnecting one link, communication remains active. This makes it more reliable than basic star or hub topologies. Therefore, redundant switch designs are preferred in professional network environments.